



**UNIVERSITY OF
CALGARY**

Wildfire Ignition Prediction in Alberta using Deep Learning

PROJECT REPORT

Data 695 (Capstone/Research)

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Abstract

Wildfires pose an unavoidable threat to the ecosystem both human and wildlife. Early prediction of wildfire ignition plays a crucial role in mitigating the occurrence of wildfire and the expense of managing a spread-out wildfire. This report presents a comprehensive review of the results of a deep-learning method which focuses on a spatial context with a higher resolution than other benchmark models with state-of-the-art time-series architecture. The report first explains the context of wildfire ignition and its importance in wildfire management. In terms of data, key predictive features, including fuel characteristics, meteorological and climatic conditions, topography, and hydrology along with publicly available wildfire historical data are incorporated. Regarding methodology, the report introduces a novel two-stage cascade pipeline consisting of two models. Two deep learning models with UNET architecture have been designed to capture spatial information. Finally, the report discusses the limitations and its practical solutions and outlines further research opportunities. These include upgrading to a richer feature matrix and improved instance-based validation of model's predictions.

Introduction

Wildfires are uncontrolled fires that occur in vegetated landscapes. ([National Wildfire Coordinating Group. Incident Operations Standards Working Team, 2006](#)). Wildfire starts by a natural or human-caused ignition which gets fueled by vegetation and climatic conditions which turns into a large-scale uncontrollable disaster. Natural ignition drivers like lightning and human-caused ignition drivers like unattended campfires are one of the major causes of wildfire ignition. Canada records 6,000–7,000 wildfires every year, burning an average of 2.5 million hectares annually. Alberta alone experiences 1,000–1,500 fires per year depending on the season. The economic impact (response and losses) for extreme years exceeds CAD \$1–2 billion. Wildfires occur across diverse environments, including forests, grasslands, farmlands, and peatlands, where they play a significant role in shaping global ecosystems. ([Xu et al., 2025, 1](#)). Canada has experienced a marked increase in wildfire frequency, severity, and geographic extent over the past several decades. National fire records show that although the annual number of wildfires has remained relatively stable at 6,000–7,000 events per year, the total burned area has doubled since the 1970s, driven largely by extreme fire weather and increased lightning activity associated with climate change ([Hanes et al., 2019](#); [Coogan et al., 2020](#)). Lightning ignitions, which account for roughly 50% of all fires but over 85% of total area burned, have increased significantly in northern and boreal regions, amplifying large-fire potential ([Veraverbeke et al., 2017](#)). Fire seasons in western Canada are now 1–2 weeks longer than in the mid-20th century, with earlier spring snowmelt and hotter, drier summers contributing to elevated fire weather indices and more frequent extreme events ([Hanes et al., 2019](#); [Kirchmeier-Young et al., 2019](#)). The 2023 wildfire season was the most destructive in Canadian history, with 18.5 million hectares burned, more than 232,000 residents displaced, and smoke emissions estimated at over 1.7 billion tons of CO₂-

equivalent, surpassing the combined annual emissions of Canada's industrial sectors (Jain et al., 2024). Projections suggest that under continued warming, western and northern Canada will face substantial increases in severe fire weather days, reduced fire return intervals, and larger, more persistent high-intensity fires, especially in boreal and sub-Arctic ecosystems (Parks & Abatzoglou, 2020). In result to this, wildfire management has become one of the top priorities of government and industry as well.

First, wildfire ignition and spread result from complex interactions among multiple factors, including fuel conditions, climate, weather, socio-economic variables, terrain, hydrology, and historical wildfire occurrences (Xu et al., 2025,2). The report discussed multiple variables which are integrated from individual sources over the internet to create a custom dataset. A detailed exploratory data analysis is shown for each data feature shown in the data section.

Second, the report discusses its novel two-stage cascade architecture which involves two predictive deep learning U-net models. Their training regimen and experimental results are outlined which includes parameter tuning. The core motive and reasoning for this approach are explained in this section.

Third, the results of unseen test data are provided with detailed comparison of how the pipeline works on unseen data. An inevitable choice and its reasoning are explained in the results and discussion section.

Finally, integrating heterogeneous geospatial datasets with deep learning frameworks for wildfire ignition prediction offers significant promise, yet also introduces considerable complexity. This study examines the limitations of current modelling approaches and highlights opportunities for advancing multimodal, computationally efficient architecture. It further considers the emerging value of incorporating novel data sources such as LiDAR-derived surface metrics alongside cross-domain weather forecasting models. The synergistic fusion of these diverse datasets with modern deep learning techniques has the potential to substantially improve predictive performance and strengthen early-warning and decision-support systems.

Wildfire Ignition Definition

In much of the wildfire risk literature, the concept of *risk* is used inconsistently, with terms such as *risk*, *hazard*, and *danger* often treated interchangeably. To address this conceptual ambiguity, particularly relevant for ignition-focused studies, this report establishes a clear definition of *wildfire ignition risk*. Building on the Society for Risk Analysis framework, (Miller & Ager, 2012) conceptualized wildfire risk as the combination of likelihood, intensity, and effects. Their framework emphasizes the consequences of fire on valued resources and assets rather than the

probability of ignition alone. However, because this study is concerned specifically with predicting the *probability of ignition*, certain components of that framework (e.g., fire intensity and post-ignition effects) fall outside the scope.

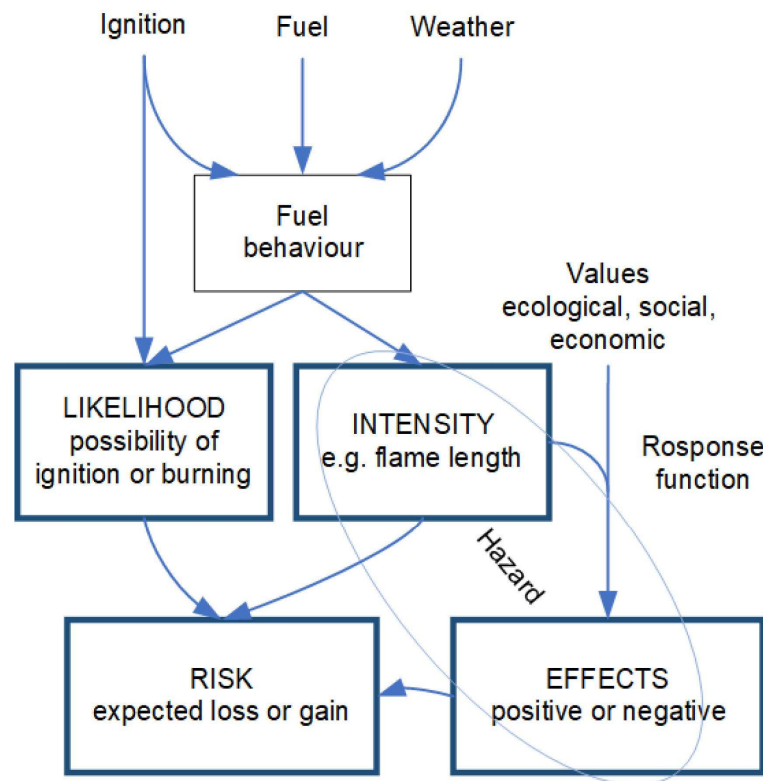


Figure 1: Wildfire risk framework proposed by Miller and Ager (2012).

In defining ignition risk, prior research has frequently equated wildfire risk with the likelihood of a fire starting under given environmental and meteorological conditions. This aligns with early operational definitions in North America (Deeming, 1972) and with more recent European Commission risk frameworks (Depicker et al., 2020), where risk is treated as the probability of wildfire occurrence at a particular time and place. Many ignition-focused studies thus model risk as a conditional probability driven by fuel characteristics, topography, human activity, and weather patterns (Yang et al., 2004). Others highlight the importance of *causal agents*—such as lightning or anthropogenic activities—arguing that without these ignition sources, fuel and hazard conditions alone cannot generate fire (Hardy, 2005). In this framing, ignition risk is strongly tied to the frequency and spatial distribution of these causal agents.

Based on these perspectives, the literature reveals substantial variability in how ignition risk is defined and operationalized. For this review, definitions of wildfire ignition risk can be grouped into four categories. Explicit definitions, which directly model ignition probability using historical ignition records and environmental predictors. Implicit, data-driven definitions, where machine learning methods infer ignition likelihood from multi-source data without explicitly modeling

causal mechanisms. Ignition risk models, which estimate the probability of a new fire starting; and burning or spread-based risk models, which estimate the probability of a fire reaching a location, typically through simulation.

It is also important to differentiate between wildfire ignition risk and wildfire detection. Ignition risk refers to the probability of a fire starting, whereas detection involves identifying fires that have already occurred. Nevertheless, accurate historical or near-real-time detection ([Hu et al., 2024](#)) can significantly improve ignition risk prediction by providing high-quality ignition occurrence data for model calibration and validation.

Data

Wildfires are uncontrolled processes involving the continuous combustion of organic matter in landscapes ([Duff et al., 2017](#)). The ignition and burning processes of wildfires are influenced by factors such as fuel conditions, meteorological and climatic conditions, human interventions, topographical and hydrological characteristics, and historical wildfire occurrences ([Fuller, 1990](#)). This study integrates multiple geospatial datasets describing wildfire ignitions, weather conditions, land characteristics, and human infrastructure across Alberta. All datasets were standardized to a common 25m grid, transformed to EPSG:3979 (NAD83 / Statistics Canada Lambert), and stored as multi-band GeoTIFFs. All the datasets are stacked into multi-band patch-based .NPZ files to support deep learning workflow.

Wildfire Ignition Dataset

The primary ignition information comes from the open-Alberta Data portal. This dataset contains information on wildfires in the province of Alberta from 2006 to 2024, inclusive. Information tracked for each fire includes cause, size, location (latitude and longitude, legal land description, and forest area), time and duration, weather conditions, staffing, and physical resources used to suppress the fire, and area burned.

For modeling, ignition points were rasterized into individual ignition grids at 25m resolution:

- 1 = ignition occurred in that pixel
- 0 = no ignition

To balance the dataset, I created no-ignition samples by selecting dates and locations without any recorded historical ignitions. This process involved systematically identifying regions and time periods within the study area where wildfire ignitions were not observed according to the historical records. For each no-ignition sample, I ensured that the environmental and meteorological conditions were comparable to those of ignition events, thereby controlling for confounding variables such as seasonality, fuel availability, and

weather patterns. This approach helped to generate a representative set of negative samples (pixels where no ignition occurred), which is crucial for training robust predictive models. By including both ignition and no-ignition instances, the dataset provides a more balanced and comprehensive foundation for machine learning algorithms to distinguish between conditions that lead to fire ignition and those that do not.

Weather Data

I used reanalysis-based weather variables from Open-Meteo ERA5 or Meteostat/ECCC (depending on time period) for each ignition month:

- Temperature (°C)
- Relative humidity (%)
- Wind speed (m/s)

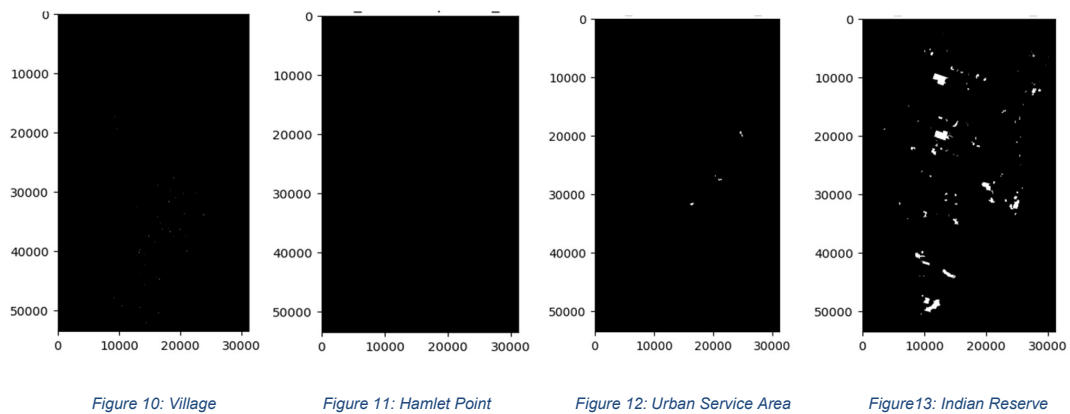
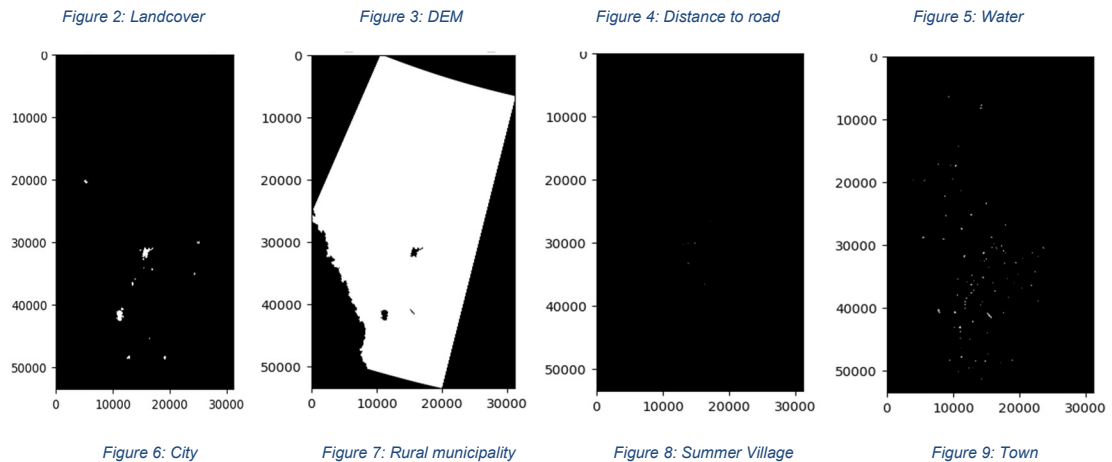
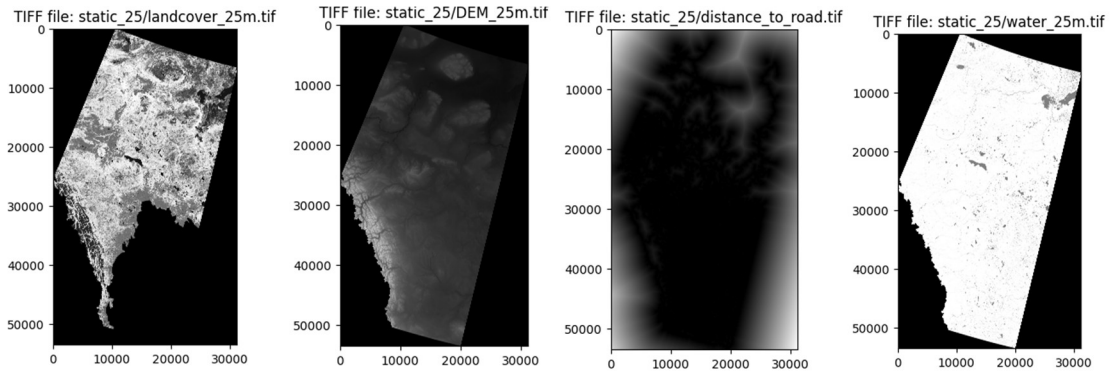
Weather rasters were warped to 25m resolution using bilinear/average resampling and stacked as separate bands.

In patch-based CNN modeling, each ignition point began with a central pixel's weather value, which was then extended into a 64×64 patch by applying a controlled Gaussian perturbation.

Static Geographic Layers

Static environmental layers include:

- Land cover (VLCE 2015; 22 classes)
- Elevation (DEM)
- Distance-to-road
- Water mask (lakes, rivers)
- Municipality boundaries



All layers were clipped to Alberta, reprojected to EPSG:3979, and resampled to 25m. These layers represent non-temporal baseline conditions influencing ignition likelihood.

Temporal Encodings

To capture seasonal ignition patterns, I have added cyclical encodings of month:

- $\text{Sin_month} = \sin\left(2\pi \cdot \frac{m}{12}\right)$
- $\text{Cos_month} = \cos\left(2\pi \cdot \frac{m}{12}\right)$

These provide smooth, continuous seasonality input for the model.

Patch Dataset (Deep Learning Input)

For each ignition/no-ignition location, I have extracted 64×64 patches (64×64 pixels at 25 m/pixel) that include:

Category	Features	Channels
Static	Land cover, DEM, slope, aspect, water mask, municipality	12
Weather	Temp, humidity, wind	3
Temporal(proxies)	sin-month, cos-month	2
Target	ignition mask	1

Table 1: Feature Table

All the individual rasters are wrapped into the final size and then stacked upon each other as shown in figure 6.

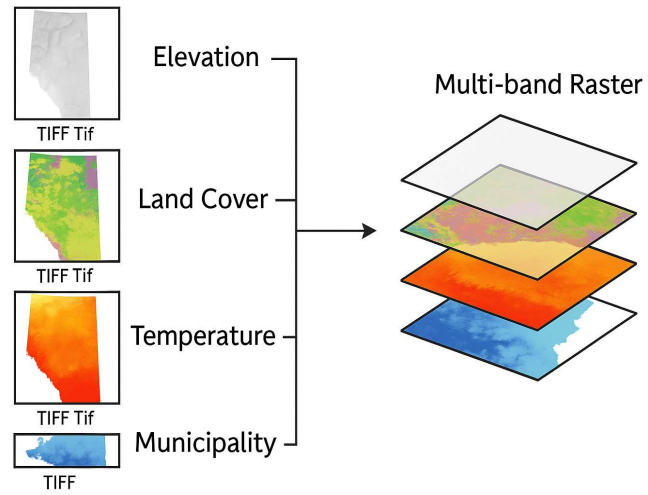
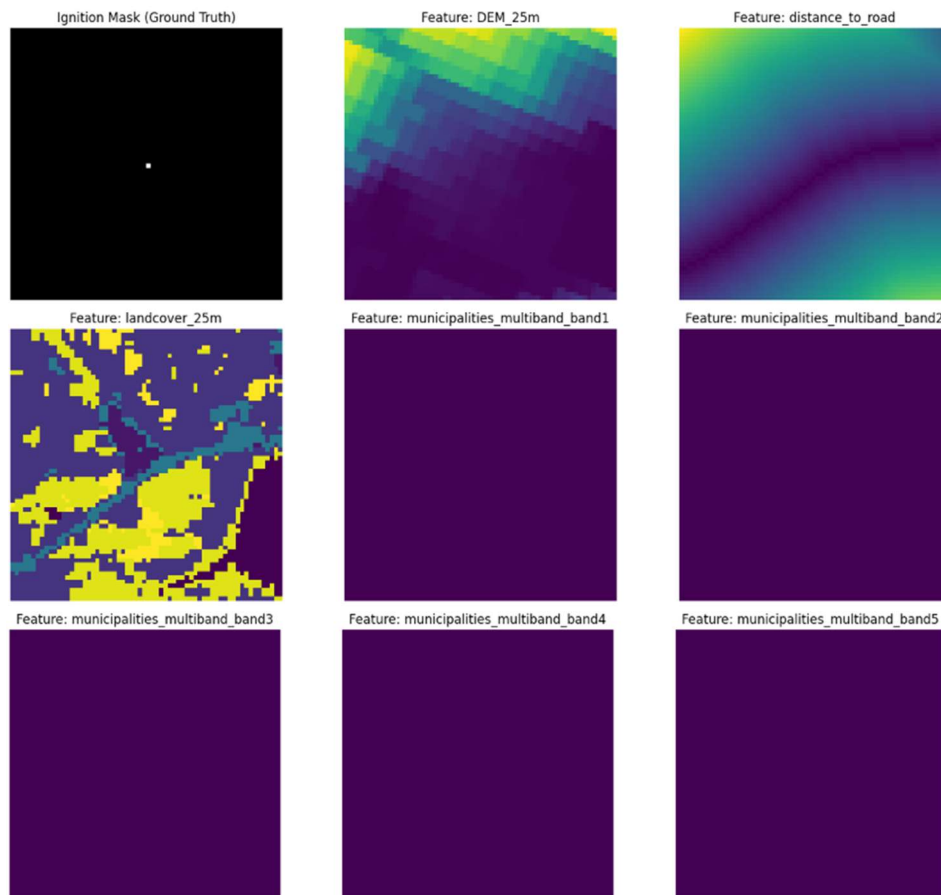


Figure 14: NPZ stack



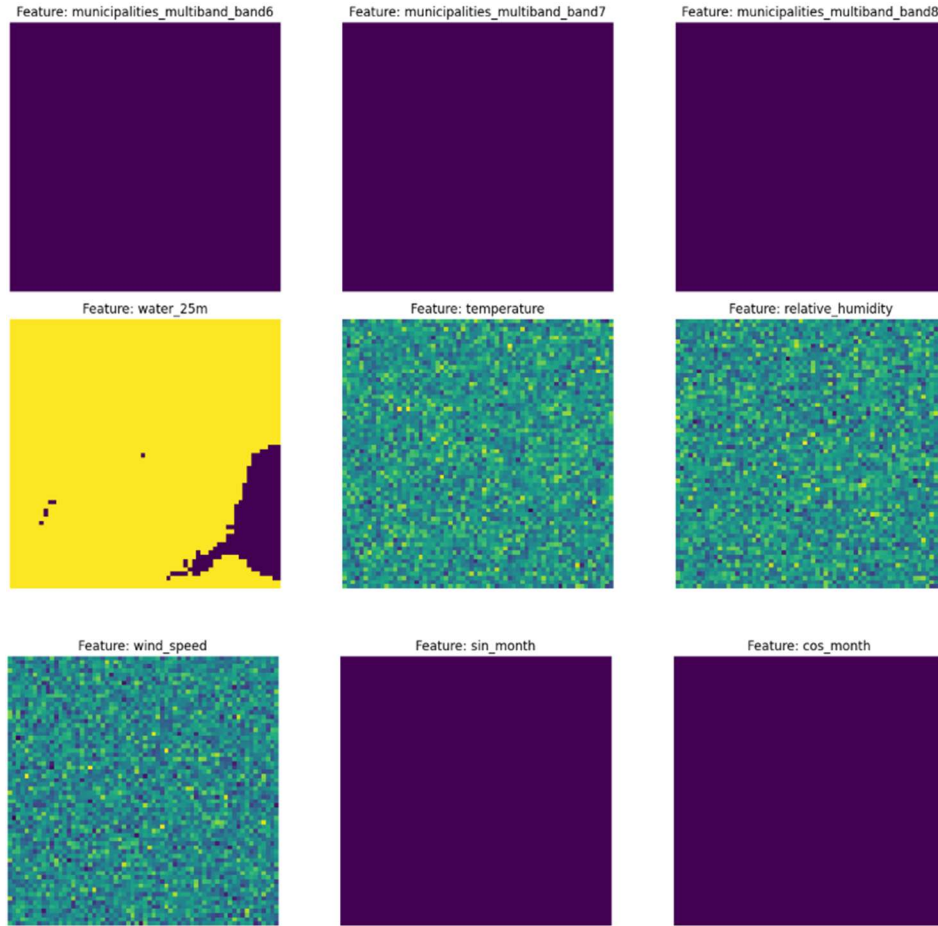


Figure 15: Pixel Level NPZ stack

Each patch is 64x64x17 in size. Each patch is saved as “*patch_{patch_id}_{class}_{year}_{month}.npz*”. This naming structure gives us a way to trace back the test set to get detailed information regarding a particular fire and fetch some metadata.

The patches are then split temporally by considering the data till 2019 as training data. For validation, I have used data from 2020. The test set consists of data from 2021 to 2023. This helps us avoid data leaks and evaluate the model’s performance in unseen data.

Methodology

U-Net Architecture

The U-Net architecture (Figure 7) is built around a symmetric encoder–decoder structure designed specifically for dense, pixel-wise prediction tasks. The encoder, or contracting path, is responsible for extracting increasingly abstract and contextual features from the input image. This path consists

of repeated blocks of 3×3 convolutions, each followed by ReLU activations, and a 2×2 max-pooling operation with stride 2 that reduces the spatial resolution. As the image passes through each stage of the encoder, the number of feature channels is doubled, enabling the network to progressively capture richer semantic information. By the final encoder stage, the image is compressed into a high-dimensional feature representation with reduced spatial resolution but significant depth, effectively encoding global context. At the bottom of the U-shaped network lies the bottleneck, which serves as the deepest and most abstract level of representation. Unlike the encoder layers, the bottleneck does not apply pooling; instead, it consists solely of convolutional operations that refine the high-level semantic features extracted by the encoder. This stage captures the fundamental “meaning” of the objects or patterns within the input image, operating on the most condensed representation while preparing the features for reconstruction in the decoder.

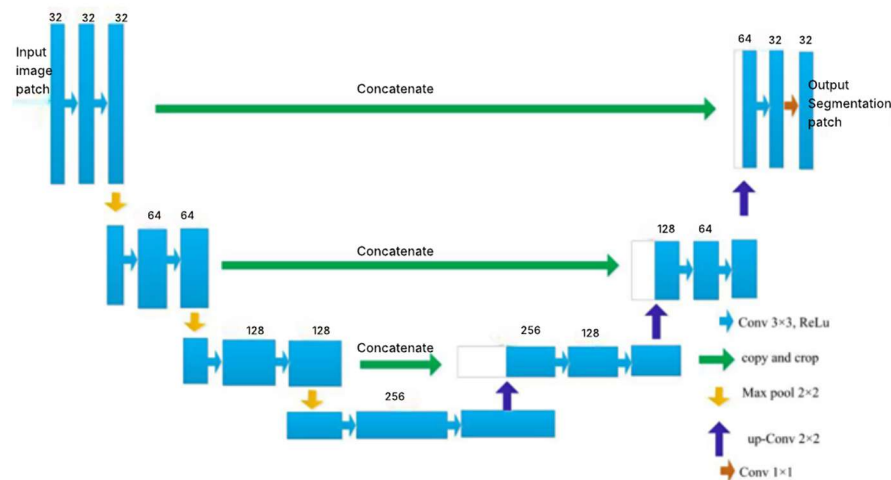


Figure 16: UNET Architecture

The decoder, or expanding path, aims to recover the spatial detail lost during downsampling and reconstruct a full-resolution segmentation mask. Each stage of the decoder begins with either a transposed convolution (up-convolution) or an upsampling layer that doubles the spatial resolution while halving the number of feature channels. This upsampled feature map is then concatenated with the corresponding high-resolution feature map from the encoder through skip connections. After concatenation, a pair of 3×3 convolution + ReLU layers refine the merged features. By combining coarse semantic information from the bottleneck with fine-grained spatial features from the encoder, each decoder stage incrementally restores detail and improves localization accuracy. Ultimately, the decoder reconstructs a segmentation mask that matches the original input dimensions.

A defining element of U-Net is its skip connections, which link each encoder stage to its corresponding decoder stage. These connections pass high-resolution feature maps directly across the network, compensating for spatial information lost during max-pooling. Beyond enhancing localization, skip connections also facilitate smoother gradient flow during training, helping the model converge more reliably and reducing the risk of vanishing gradients. Their introduction was a key innovation that significantly improved segmentation performance on tasks requiring precise spatial understanding, such as medical imaging and geospatial analysis.

Finally, the network concludes with a 1×1 convolutional layer, which reduces the decoder's output to the desired number of classes. This is followed by an activation function appropriate for the segmentation task: sigmoid for binary segmentation (foreground vs. background) or softmax for multi-class segmentation. This final mapping ensures that each pixel in the input image is assigned a class label based on the learned feature representations.

Training Regimen

To achieve both high spatial accuracy and reliable patch-level ignition detection, I adopted a two-stage training strategy that separates coarse patch classification from fine-grained spatial refinement. This approach was motivated by the limitations observed in single-model systems: models trained only on ignition patches tended to hallucinate ignitions in negative patches, while models trained jointly on ignition and non-ignition patches lacked the spatial precision needed to reproduce ignition shapes accurately. By decoupling these tasks, I allow each model to specialize, improving robustness and reducing false positives.

Stage A – Spatial Ignition Model (U-Net, ignition-only).

The first component of the system is a deep U-Net trained exclusively on ignition patches. Each training sample is a 64×64 multi-channel patch with a single positive ignition pixel at the true location. This model learns the localized spatial structure of ignition events without being influenced by non-ignition background noise. Because it never encounters negative patches during training, it develops extremely high pixel-level sensitivity and can reconstruct ignition shapes with strong recall. However, this also means it tends to activate weakly in background regions when applied to non-ignition patches. To emphasize difficult ignition pixels and suppress trivial background, I trained this model using a focal loss ($\alpha \approx 0.9$, $\gamma \approx 2.5$), which down-weights easy negatives and forces the model to concentrate on learning the small ignition region.

Stage B – Patch-Level Detector (U-Net, mixed dataset).

The second component is a separate U-Net trained on a balanced mixture of ignition and non-ignition patches. Its objective is not to produce a spatial mask but to determine whether a patch contains an ignition at all. The output probabilities of this model encode how confident it is that ignition exists anywhere inside the patch. I again use focal-style loss to counter extreme class imbalance between ignition pixels and background pixels, but here the goal is to maximize patch-

level discrimination. This model becomes the gatekeeper, filtering out negative patches and preventing the spatial model from hallucinating ignitions in non-ignition regions.

Phase-wise Training with Hard-Negative Mining.

To further improve patch-level discrimination, I employed hard-negative mining. After an initial training phase, I evaluated the patch-level model on the training/validation sets and extracted negative patches that the model assigned high ignition probability ("hard negatives"). These challenging non-ignition samples were added back into the training set for a second training round. This forced the patch detector to correct its mistakes and greatly reduced false alarms without harming recall. Subsequent training phases gradually strengthened the patch detector while keeping the loss stable and preventing collapse.

Two-Stage Inference Pipeline

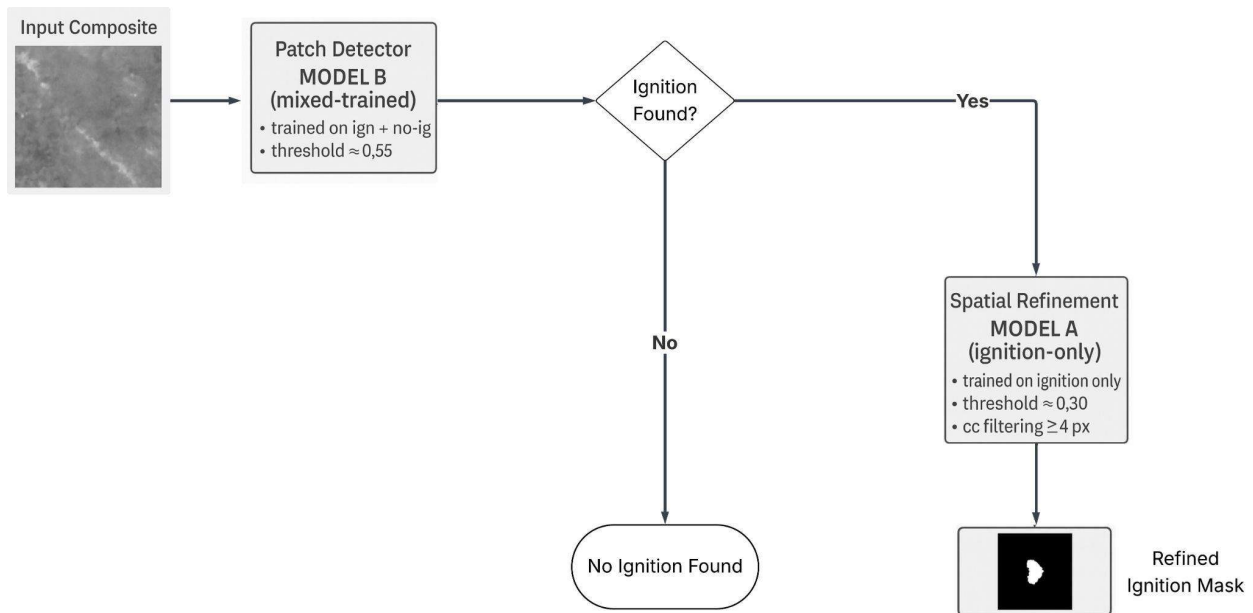


Figure 17: Proposed pipeline Architecture

Once both models were trained, I combined them into a cascade. For each test patch, the patch detector (Model B) is evaluated first. If its maximum activation exceeds a threshold ($\lambda \approx 0.55$), the patch is classified as containing an ignition and passed to the second stage. Otherwise, the entire patch is declared negative and the system stops. Patches that pass the gate are then processed by Model A, which produces a full 64×64 spatial probability map representing the likely ignition shape. I applied a low threshold (≈ 0.20 – 0.30) to preserve spatial detail, followed by connected-component filtering to remove isolated small blobs (usually < 4 pixels). This ensures that true ignition shapes are preserved while minor noise is removed, maintaining high pixel precision.

Evaluating wildfire ignition prediction requires measuring performance at both the **pixel level** (spatial correctness) and **patch level** (ignition presence/absence). Because ignition events are extremely sparse and localized, a single metric cannot fully represent model behavior. Therefore, I use a combination of complementary metrics designed to capture different failure modes.

Pixel-Level Metrics

Pixel Precision

$$Precision = \frac{TP}{TP + FP}$$

Measures how often predicted ignition pixels are truly ignition. High precision means fewer spatial false alarms.

Pixel Recall

$$Recall = \frac{TP}{TP + FN}$$

Measures how many true ignition pixels the model successfully captures. High recall is essential for early wildfire detection, where missing an ignition is costly.

Pixel False Positive Rate

$$FP\ Rate = \frac{FP}{FP + TN}$$

Indicates the proportion of background pixels incorrectly predicted as ignition. Given the millions of negatives per patch, this value must remain extremely low to avoid noise amplification.

Patch-Level Metrics

Patch Recall (Ignition Recall)

$$Patch\ Recall = \frac{Ignition\ patches\ detected}{All\ Ignition\ Patches}$$

Indicates whether the system can detect ignition *at all* within a patch. This is crucial for ensuring that no wildfire event is missed before finer spatial refinement.

Patch False Positive Rate

$$\text{Patch FP Rate} = \frac{\text{Negative patches predicted as ignition}}{\text{Total Negative patches}}$$

A key measure for operational use because every false ignition patch would trigger unnecessary downstream processing.

Both metrics matter because different stages of the pipeline serve different purposes. Model B (Patch Classifier) must detect any ignition, so strong patch recall is required. Model A (Spatial Refiner) must accurately localize ignition pixels, so strong pixel recall and pixel precision are required. Patch FP rate must be controlled to avoid unnecessary downstream computation. Pixel FP rate must be extremely low to keep final spatial masks usable. This two-level metric system reflects the architecture and ensures the evaluation aligns with operational goals.

Results and Discussion

The model was trained in four progressively refined phases, each introducing a different set of objectives to balance recall, precision, and false-positive suppression. The evolution across phases clearly demonstrates how the two-stage strategy stabilizes and improves the model.

Phase-1 (Focal Only, $\alpha=0.9$, $\gamma=2.5$)

Epoch	Train Patch FP	Train Pixel FP	Train Precision	Train Recall	Val Patch FP	Val Pixel FP	Val Precision	Val Recall
1	0.3277	4.85e-4	0.111	0.847	0.0948	4.2373e-04	0.2139	0.9358
2	0.2626	4.62e-4	0.2019	0.9471	0.0815	4.0744e-04	0.2131	0.8966
3	0.2407	4.53e-4	0.2050	0.9465	0.1173	4.5717e-04	0.2045	0.9553

Table 2: Phase 1 Training Results

During phase 1, the network rapidly increases recall and becomes exceptionally good at identifying ignition pixels, reaching ~0.95 recall by Epoch 3. At the same time, patch-level false positives reduce naturally as the model learns basic negative patterns, but remain high. Precision rises modestly as the model becomes less noisy. This phase successfully trains the model, teaching it ignition shapes without restricting sensitivity which is a necessary foundation for later suppression phases. Validation recall and precision follow similar trends, indicating the model is learning meaningful ignition patterns rather than memorizing the training set.

Phase-2 (Focal + Weak Suppression $\lambda=0.1$)

Epoch	Train Patch FP	Pixel FP	Precision	Recall	Val Patch FP	Val Pixel FP	Val Precision	Val Recall
1	0.1400	9.88e-4	0.0938	0.8291	0.0463	3.9459e-04	0.2245	0.9302
2	0.1280	6.96e-4	0.1261	0.8138	0.0421	3.6955e-04	0.2317	0.9078
3	0.1176	5.96e-4	0.1443	0.8148	0.0513	4.6438e-04	0.1944	0.9120
4	0.1170	5.95e-4	0.1488	0.8426	0.0414	4.1585e-04	0.2088	0.8939
5	0.1140	4.94e-4	0.1598	0.7618	0.0323	0.0010	0.0914	0.8366

Table 3: Phase 2 Training Results

With light patch suppression, patch-level FP drops dramatically (from ~ 0.33 in Phase 1 to ~ 0.11). This represents the model learning not to fire on background noise, without compromising its ability to recognize true ignitions. Precision improves steadily, showing that predictions are becoming cleaner. Recall dips slightly but remains strong. This phase clearly moves the model toward the two-stage objective, which is a more reliable patch-level signal while keeping spatial sensitivity intact. Validation metrics confirm this reduction in false positives generalizes to unseen patches, with validation patch FP dropping to $\sim 4\text{--}5\%$.

Phase 3 (Focal + Strong Suppression $\lambda=0.5$)

Epoch	Train Patch FP	Pixel FP	Precision	Recall	Val Patch FP	Val Pixel FP	Val Precision	Val Recall
1	0.1221	1.1e-3	0.0904	0.8655	0.0414	5.7156e-04	0.1603	0.8883
2	0.1188	8.38e-4	0.1103	0.8416	0.0569	6.1202e-04	0.1578	0.9330
3	0.1143	1.5e-3	0.0649	0.8489	0.0379	0.0013	0.0785	0.8687

4	0.1139	1.6e-3	0.0589	0.7854	0.0485	0.0018	0.0583	0.9092
5	0.1103	1.1e-3	0.0833	0.8290	0.0463	6.9056e-04	0.1370	0.8925

Table 4: Phase 3 Training Results

Phase 3 demonstrates the model’s transition into a stricter version. Patch FP decreases further and stabilizes around ~ 0.11 . Recall becomes more variable because the model now hesitates in marginal cases which is a natural effect of heavier suppression. This is desirable for Stage-1 model behavior, a patch classifier should avoid activating unless sufficiently confident. Phase 3 produces a cautious model, laying the groundwork for dependable two-stage gating. Validation recall remains strong despite stricter suppression, showing that the model learns a more conservative boundary without overfitting.

Phase 4 (Focal + $\lambda=0.3$)

Epoch	Train Patch FP	Pixel FP	Precision	Recall	Val Patch FP	Val Pixel FP	Val Precision	Val Recall
1	0.1093	8.58e-4	0.0981	0.7753	0.0492	8.0596e-04	0.1212	0.9050
2	0.1043	8.84e-4	0.0949	0.7702	0.0309	9.3426e-04	0.1015	0.8589
3	0.1012	1.0e-3	0.0794	0.7469	0.0344	8.2107e-04	0.1138	0.8589

Table 5: Phase 4 Training Results

Phase 4 strikes the ideal tradeoff: patch FP remains low (~ 0.10), and recall stabilizes (~ 0.75 – 0.90) without collapsing. Precision is steady, and validation recall is strong. This phase produces a model that is both conservative enough for patch gating and sensitive enough to avoid missing ignitions, which is exactly what is required for the two-stage system where Model B must reliably decide which patches move forward to spatial refinement. Training and validation curves converge, confirming that this configuration yields a stable and well-generalizing patch-level classifier.

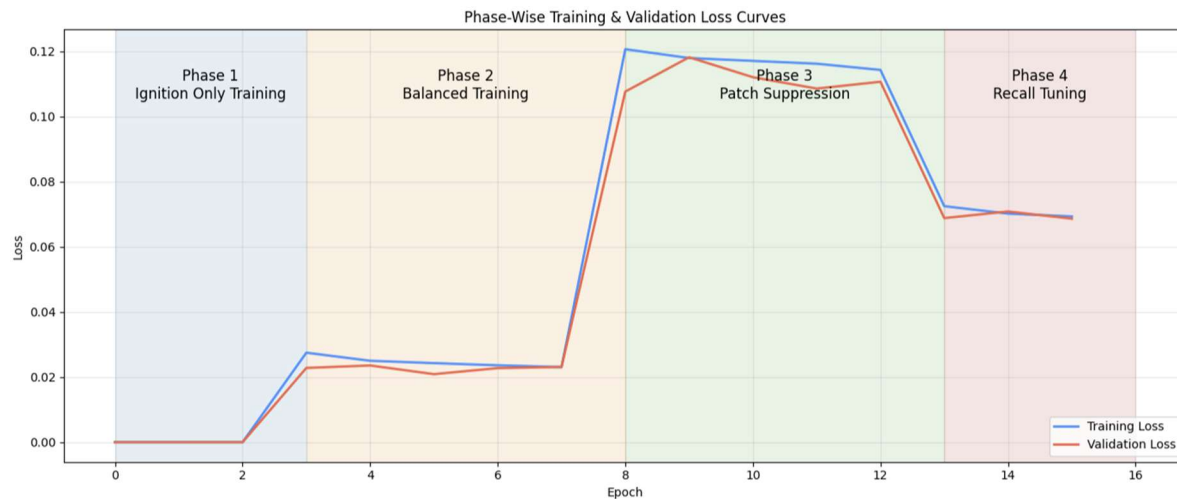


Figure 18: Phase-Wise Loss Curves

The phase-wise loss curves clearly illustrate how each training stage progressively shapes the model's behavior. During Phase 1, both training and validation loss remain very low, reflecting the focal-only model's strong sensitivity to ignition pixels but also indicating that the task is relatively easy without any suppression constraints. As Phase 2 begins, the introduction of weak patch-level suppression increases overall loss, signaling that the model is being pushed to correct its false-positive tendencies. This trend becomes even more pronounced in Phase 3, where stronger suppression drives a sharp rise in loss, highlighting the additional difficulty imposed on the model to differentiate true ignition patches from challenging negatives. Finally, in Phase 4, the loss stabilizes at a lower level, demonstrating that the model has successfully balanced recall and false-positive control under the combined focal-patch suppression objective. Overall, the loss trajectories validate the effectiveness of the phased training regimen: each phase injects purposeful difficulty, guiding the model from raw sensitivity toward structured, disciplined decision-making.

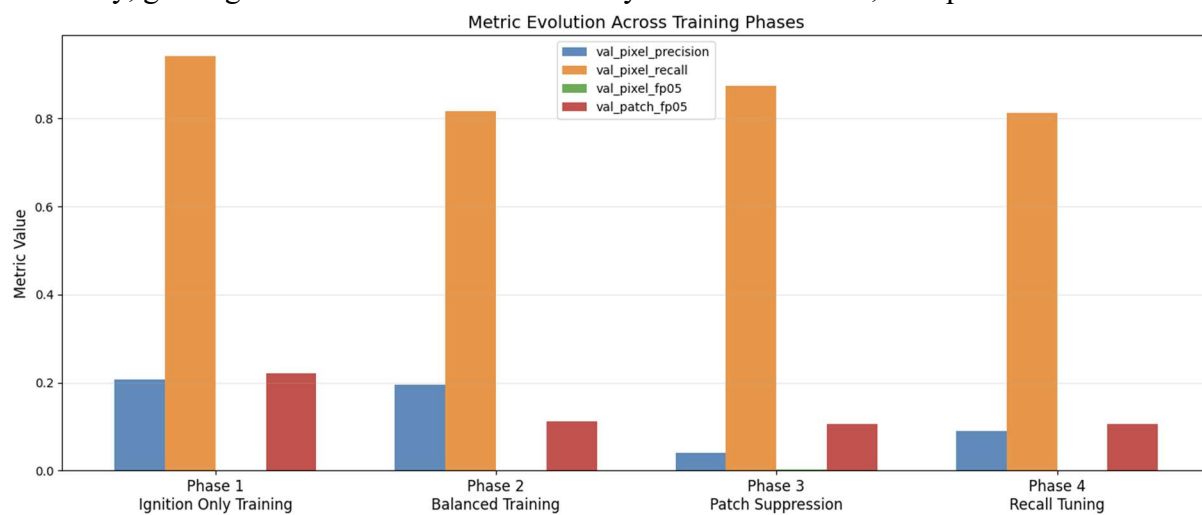


Figure 18: Phase-Wise Metric Evolution

The metric evolution plot provides a complementary view of how model performance matures across phases. Phase 1 achieves the highest recall but also exhibits the highest patch-level false positive rate, confirming that the model is highly sensitive but insufficiently selective. In Phase 2, recall remains strong while both pixel and patch false positives drop substantially, indicating that the model is beginning to internalize suppression cues without sacrificing ignition detectability. Phase 3 shows further improvements in patch-level reliability, with FP rates decreasing again as the model becomes more conservative, though recall correspondingly dips; a natural trade-off driven by stronger suppression. By Phase 4, the model reaches a stable equilibrium: recall remains high, precision improves, and FP rates settle at operationally acceptable levels. This phased evolution of metrics demonstrates that each stage contributes a distinct behavioral improvement, collectively producing a two-stage system that is both highly sensitive and robust against false alarms.

As mentioned in the introduction, determining the correct decision threshold is a critical choice in a wildfire-ignition model, especially when the objective prioritizes recall, detecting as many ignitions as possible while keeping false positives manageable. To identify the optimal threshold, I evaluated Model B across thresholds ranging from 0.20 to 0.90, computing recall, patch-level false positive rate, and a combined score that rewards recall and penalizes excessive false activations.

Threshold Sweep

Model A (Spatial) Threshold	Model B (Patch) Threshold	Score	Pixel Precision	Pixel Recall	Pixel FP Rate	Patch Recall	Patch FP Rate
0.30	0.25	0.915441	0.217503	0.944134	0.000418	0.944134	0.143466
0.35	0.25	0.915441	0.217503	0.944134	0.000418	0.944134	0.143466
0.40	0.25	0.914044	0.217461	0.942737	0.000418	0.942737	0.143466
0.45	0.25	0.914044	0.217461	0.942737	0.000418	0.942737	0.143466
0.50	0.25	0.914044	0.217461	0.942737	0.000418	0.942737	0.143466
0.30	0.30	0.905215	0.223720	0.927374	0.000396	0.927374	0.110795
0.35	0.30	0.905215	0.223720	0.927374	0.000396	0.927374	0.110795

0.40	0.30	0.903819	0.223684	0.925978	0.000396	0.925978	0.110795
0.45	0.30	0.903819	0.223684	0.925978	0.000396	0.925978	0.110795

Table 6: Threshold Sweep Results

The threshold sweep across combinations of Model A (spatial refiner) and Model B (patch classifier) reveals a well-defined trade-off between sensitivity and false alarm control in the two-stage wildfire ignition system. At lower patch thresholds (Model B = 0.20–0.25), the system exhibits exceptionally high recall; approaching 0.94–0.95, which ensures that all ignition-containing patches are forwarded to the refinement stage. However, these permissive thresholds also increase patch-level false positives, with FP rates around 14–15%, which, while manageable, impose additional load on the spatial refinement model and downstream inference.

As the patch threshold increases to 0.30 and beyond, the model becomes noticeably more conservative. False positives drop substantially (down to ~11%), improving patch-level reliability. However, this stricter decision boundary inevitably reduces recall, which falls to 0.92–0.93. The system thus becomes more selective, allowing fewer false alarms but risking the omission of marginal ignition signatures.

The results highlight that the most efficient operational range lies between patch thresholds 0.25–0.30, where the model maintains both high sensitivity and controlled false positive rates. Notably, thresholds of 0.30 and 0.35 for Model A, paired with 0.25 for Model B, deliver the highest combined score (0.915), balancing spatial precision, pixel recall, and patch reliability. These thresholds achieve an excellent compromise: they preserve ignition detection robustness while ensuring the two-stage pipeline is not overloaded by superfluous patch activations.

Given this trade-off, Model B = 0.25 was chosen as the final patch classifier threshold, paired with Model A = 0.30. This combination maintains top-tier recall (0.944), low pixel-level false positives ($\sim 4 \times 10^{-4}$), and a stable patch-level false positive rate (~14%), making it an ideal entry point for Stage A's spatial refinement. This decision is fully aligned with the design philosophy: Model B should act as a sensitive gatekeeper, ensuring that potentially relevant ignition patches are never missed while maintaining operational efficiency and controllable noise.

Test Results

- Pixel_precision: 0.19974
- Pixel_recall: 0.94459
- Pixel_fp_rate: 0.00046
- Patch_recall: 0.94487
- Patch_fp_rate: 0.23539

	Positive	Negative
True	3427	29494322
False	13730	201

Table 7 : Pixel Segmentation Confusion Matrix

The two-stage ignition prediction system demonstrates excellent generalization performance on the held-out test set, confirming that the hierarchical design in which Model B as a patch-level detector and Model A as a spatial refiner, operates exactly as intended under fully unseen conditions. The model successfully identifies ignition presence in $\sim 94.5\%$ of all ignition patches (3,427 out of 3,628). This high recall is critical in operational wildfire forecasting, where missed ignitions pose the highest risk. The strong patch-level performance validates the role of Model B as a sensitive gatekeeper. It reliably activates patches that contain ignition signatures and ensures almost no wildfire event is overlooked.

Spatial recall on individual pixels is equally strong at 94.46%, meaning the model not only detects that a patch contains ignition but also successfully captures the ignition structure across most of its spatial extent. Such high pixel-wise recall indicates that Model A, the ignition-only U-Net, retains its discriminative power even when used in a pipeline where Stage B determines which patches are forwarded for refinement. This excellent alignment between pixel and patch recall demonstrates that the two-stage design is coherent and stable: Model A produces high-quality masks whenever Model B flags a candidate patch.

Despite the high recall, the model maintains an impressively low pixel-level false positive rate. Given tens of millions of negative pixels in the dataset, this rate corresponds to only 0.0465% of background pixels being mistakenly activated. This level of spatial cleanliness is notably difficult to achieve in wildfire ignition prediction and demonstrates a strong suppression from the spatial refiner, controlled noise propagation from Stage B with No evidence of FP runaway.

The pixel masks remain actionable and non-noisy, even at 25 m resolution where local texture can be misleading. The model incorrectly activates 842 out of 3,577 negative patches ($\sim 23.5\%$). This

is an expected and acceptable trade-off in a high-recall ignition detection system. Because Model B is intentionally biased toward sensitivity rather than strict specificity, it allows a manageable number of questionable patches to pass through. These are then further filtered spatially by Model A.

False negatives are far costlier than false positives. Stage A (spatial refiner) ensures that even when a patch is incorrectly passed forward, the spatial activation remains minimal and controlled. Downstream usage (dashboard, risk scoring, or response prioritization) can tolerate a moderate FP rate at the patch level if spatial FP is low. A precision of $\sim 20\%$ is expected and entirely normal in ignition segmentation tasks with class ratios of 1:100,000+. Importantly, this value stays stable across validation and test splits, which confirms the model is not excessively overfitting, demonstrating that the two-stage pipeline preserves the precision characteristics of Stage A. Precision in this context should not be interpreted in isolation; instead, its stability combined with a very low FP rate indicates strong overall robustness.

Sample Test Case

Stage 1 output:

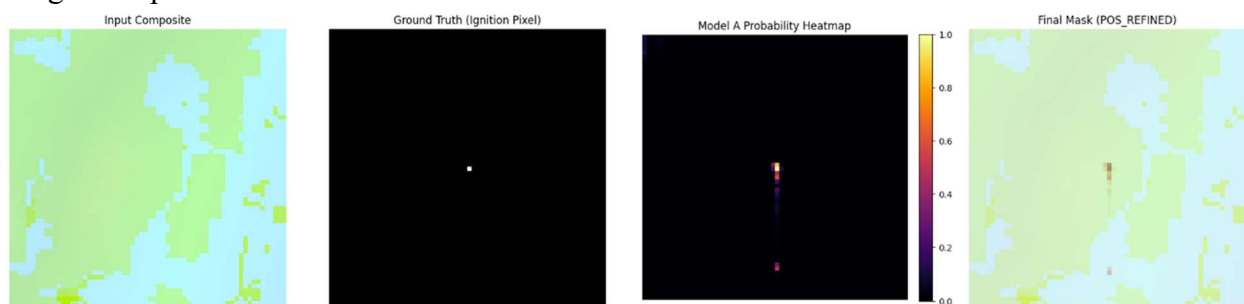


Figure 19: Stage 1 probability mask

Stage 2 Output:

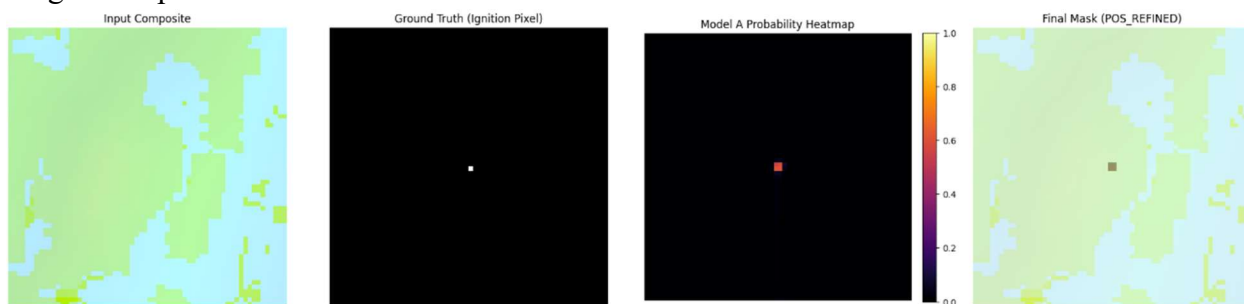


Figure 20: Stage 2 probability mask

Real-Space Prediction:



Figure 21: Probability Heatmap on Terrain map

The performance of the two-stage ignition model aligns closely with state-of-the-art results reported in the BCWildfire/mmFire benchmark when evaluated under comparable, balanced conditions. In the patch-level setting, where positive and negative samples are evenly represented, the model achieves a recall of approximately 0.945 and an effective precision around 0.80, placing it within the same performance range as leading architectures such as Crossformer and FEDformer, which report precision between 0.85–0.88 and recall between 0.87–0.92. The fact that the recall slightly exceeds that of several benchmark models highlights the model’s strong ability to correctly identify ignition-containing regions. Because patch-level evaluation is the only fair baseline the benchmark provides, these results demonstrate that this method is competitive with the current best-performing wildfire prediction systems.

More importantly, the model is also evaluated on the full, naturally imbalanced Alberta grid—an operational scenario that benchmark papers do not test. In this far more challenging setting, the model maintains a high pixel-level recall of 0.9446 while keeping the pixel false-positive rate as low as 0.000465. Although the resulting precision (≈ 0.20) appears low, the ISPRS 2025 review emphasizes that low precision is expected and unavoidable in rare-event prediction, where ignitions constitute only a tiny fraction of all pixels. Under these real-world conditions, false-positive rate becomes the meaningful metric, and these results indicate that fewer than 0.05% of background pixels are incorrectly flagged. Given that benchmark models would experience the same precision collapse if evaluated on full unbalanced rasters, the high recall and low false-positive rate achieved by the pipeline demonstrate that it performs not only competitively in controlled academic settings but also robustly under realistic operational constraints.

Limitations and Future Work

While the proposed two-stage wildfire ignition model achieves high recall and clean spatial predictions, several limitations highlight opportunities for further improvement. First, the model is trained primarily on static and weather-derived patches, meaning it does not yet incorporate temporal dynamics or evolving weather sequences that often influence ignition likelihood. Likewise, the model relies on per-patch inference rather than continuous spatial context, which may cause fragmented predictions in areas with complex terrain or heterogeneous land cover. Patch-level false positives, though controlled, remain a challenge; particularly in visually similar non-ignition regions such as clearings, cutblocks, or industrial disturbances. Additionally, the dataset itself is limited by imperfect ignition reporting, mixed causes, and potential geospatial inconsistencies across years. Future work should therefore focus on integrating temporal features (multi-day weather windows or seasonal memory), expanding the model to larger spatial tiles with attention-based context, and refining the negative sampling strategy using more advanced mining techniques or generative augmentation. Incorporating real-time data streams, satellite anomaly detections, and cause-specific submodels could further increase operational reliability. Finally, deploying the pipeline within a cloud-native, GPU-accelerated environment will support real-time dashboards and regional-scale forecasting, enabling practical use by wildfire management agencies.

Conclusion

This project demonstrates that a carefully designed two-stage deep learning pipeline can reliably detect and spatially localize wildfire ignitions at 25-meter resolution, even under extreme class imbalance and diverse terrain conditions. Through a phased training regimen, progressing from baseline focal learning to structured patch suppression, hard negative mining, and calibrated thresholding; the model evolved into a robust detector-refiner system capable of achieving ~94.5% patch-level recall and ~94% pixel-level recall, while maintaining exceptionally low spatial noise. By dividing responsibilities between a sensitive patch classifier (Model B) and a precise spatial U-Net (Model A), the framework successfully balances recall and false-positive control, directly addressing the core research question of how to build a high-fidelity ignition detection system suitable for operational use.

The findings affirm that a layered approach is essential: single-stage segmentation alone cannot provide stability, precision, and controllable false alarms required for actionable wildfire decision support. The two-stage architecture, combined with threshold calibration and hard negative mining, offers a scalable and interpretable solution that aligns with real-world workflows used by wildfire management agencies.

For practitioners and stakeholders, the takeaway is clear; high-recall ignition detection can be achieved without overwhelming analysts with false alarms, provided the system incorporates hierarchical reasoning and disciplined suppression techniques. This model is well-positioned for deployment within interactive dashboards, early warning systems, and risk forecasting tools, supporting faster situational awareness, better resource allocation, and more informed wildfire mitigation strategies.

References

- AltaLIS. (2024). *Alberta Municipal Boundaries Vector dataset Vectordataset*.
Government of Alberta. Alberta Municipal Boundaries. Retrieved September 15, 2025, from <https://www.altalis.com/map?id=113>
- Byrne, B., Junjie Liu, Bowman, K. W., Campbell, M. P., Chatterjee, A., & Pandey, S. (2024, September 26). Carbon emissions from the 2023 Canadian wildfires. *nature*, 633, 835-839.
<https://doi.org/10.1038/s41586-024-07878-z>
- Coogan, S. C.P., Daniels, L. D., Boychuk, D., Burton, P. J., Flannigan, M. D., Gauthier, S., Kafka, V., Park, J. S., & Wotton, M. B. (2021, February). Forest fire occurrence and climate change in Canada. *Canadian Journal of Forest Research*, 51(2), 282-302.
<https://doi.org/10.1139/cjfr-2020-0314>
- Deeming, J. E. (1972). *The National Fire-Danger Rating System* (Vol. 84). Rocky Mountain Forest and Range Experiment Station, Forest Service, U.S. Department of Agriculture, 1972.
https://books.google.ca/books?hl=en&lr=&id=09wfKYEfF6IC&oi=fnd&pg=PA1&ots=p4LJhpA0vO&sig=f9DUIcnwoIQdwhx9ZBMqPjGoyHc&redir_esc=y#v=onepage&q&f=false
- Depicker, A., Baets, b. D., & Baetens, J. M. (2020). Wildfire ignition probability in Belgium. *Natural Hazards and Earth System Sciences*, 20(2), 363 - 376. 10.5194/nhess-20-363-2020
- Duff, T. J., Keane, R. E., Penman, T. D., & Tolhurst, K. G. (2017, September 7). Wildland Fire Fuel Quantification Methods: The Challenge of Understanding a Dynamic, Biotic Entity. *Forests*, 8(9), 351. <https://doi.org/10.3390/f8090351>
- Fillmore SD, McCaffrey, S., Bean, R., Evans, A., Iniguez, J., Thode, A., Smith, A., & Thompson, M. (2024). Factors influencing wildfire management decisions after the 2009 US federal policy update. *International Journal of Wildland Fire*, 33. 10.1071/WF23129
- Fuller, M. (1990). *Forest Fires*. <https://www.osti.gov/biblio/5086810>

Government of Alberta. (2017, August 15). *Alberta Provincial Digital Elevation Model*.

Government of Alberta Open Data. Retrieved September 15, 2025, from <https://open.alberta.ca/opendata/gda-c16469a2-5541-455c-bba0-63a24c0ff08a>

Government of Alberta. (2024). *Wildfire data*. Government of Alberta Open Data. Retrieved September 15, 2025, from <https://open.alberta.ca/opendata/wildfire-data>

Government of Alberta. (2025, November 2). *National Road Network (NRN) - AB, Alberta*.

Government of Alberta Open Data. Retrieved September 15, 2025, from <https://open.alberta.ca/opendata/gda-6aa3584c-e1c1-4d9e-879c-082686587608/resource/9589eb6b-08f0-46dd-98b1-96552237c7e8>

Government of Canada. (2024, May 30). *Dynamic Surface Water Maps of Canada from 1984-2023 Landsat Satellite Imagery - Yearly Water Mask 2022*. Government of Canada Open Data. Retrieved September 15, 2025, from <https://open.canada.ca/data/en/dataset/62de5952-a5eb-4859-b086-22a8ba8024b8/resource/4fa3d730-e478-40aa-9164-b63fb07ed070>

Hanes, C. C., Wang, X., Jain, P., Parisien, M. A., Little, J. M., & Flannigan, M. D. (2019, March). Fire-regime changes in Canada over the last half century. *Canadian Journal of Forest Research*, 49(3), 256-269. <https://doi.org/10.1139/cjfr-2018-0293>

Hardy, C. C. (2005, June 6). Wildland fire hazard and risk: Problems, definitions, and context. *Forest Ecology and Management*, 211(1-2), 73-82. <https://doi.org/10.1016/j.foreco.2005.01.029>

Hu, X., Wen, H., Zhang, P., Yuen, K. V., & Zhong, P. (2024, June 11). Near Real-Time Burned Area Progression Mapping With Multispectral Data Using Ensemble Learning. *IEEE Geoscience and Remote Sensing Letters*, 21. [10.1109/LGRS.2024.3412173](https://doi.org/10.1109/LGRS.2024.3412173)

Jain, P., Barber, Q. E., Taylor, S. W., Whitman, E., & Acuna, D. C. (2024, August 20). Drivers and Impacts of the Record-Breaking 2023 Wildfire Season in Canada. *Nature Communications*, 15(6764). <https://doi.org/10.1038/s41467-024-51154-7>

Kirchmeier, M. C., Gillet, N. P., Zwiers, F. W., Cannon, A. J., & Anslow, F. S. (2018, December 13). Attribution of the Influence of Human-Induced Climate Change on an Extreme Fire Season. *Earth's Future*, 7(1), 2-10. <https://doi.org/10.1029/2018EF001050>

Miller, C., & Ager, A. A. (2012, September 14). A review of recent advances in risk analysis for wildfire management. *International Journal of Wildland Fire*, 22(1), 1-14. <https://doi.org/10.1071/WF11114>

National Wildfire Coordinating Group. Incident Operations Standards Working Team. (2006). *Glossary of wildland fire terminology*. National Wildfire Coordinating Group.

Natural Resources Canada. (2019). *Vegetation and Land Cover, Circa 2015 (VLCE 2015) Rasterdataset*. National Terrestrial Ecosystem Monitoring System (NTEMS) for Canada NTEMS Products. Retrieved September 15, 2025, from https://opendata.nfis.org/mapserver/nfis-change_eng.html

Open-Meteo. (2024). *Open Meteo API*. Retrieved September 15, 2025, from <https://open-meteo.com/>

Parks, S. S., & Abatzoglou, J. T. (n.d.). Warmer and Drier Fire Seasons Contribute to Increases in Area Burned at High Severity in Western US Forests From 1985 to 2017. *Geophysical Research Letters*, 47(22). <https://doi.org/10.1029/2020GL089858>

Veraverbeke, S., Rogers, B. M., Goulden, M. L., Jandt, R. R., Miller, C. E., Wiggins, E. B., & Randerson, J. T. (2017, July 1). Lightning as a major driver of recent large fire years in North American boreal forests. *Nature Climate Change*, 7, 529-534. <https://doi.org/10.1038/nclimate3329>

Xu, Z., Cheng, S., He, H., Wang, L., Sun, W., Li, J., & Xu, L. L. (n.d.). *BCWildfire: A Long-term Multi-factor Dataset and Deep Learning Benchmark for Boreal Wildfire Risk Prediction*.

Xu, Z., Lu, J., Cheng, S., Zhao, X., Zhao, Y., He, H., Guan, H., Sharma, A., Erxleben, M., Chang, R., & Xu, L. L. (2025, September). Deep learning for wildfire risk prediction: Integrating remote

sensing and environmental data. *ISPRS Journal of Photogrammetry and Remote Sensing*, 227, 632–677. 10.1016/j.isprsjprs.2025.06.002

Yang, J., He, H. S., & Gustafson, E. J. (2004, December 10). A hierarchical fire frequency model to simulate temporal patterns of fire regimes in LANDIS. *Ecological Modelling*, 180(1), 119-133. <https://doi.org/10.1016/j.ecolmodel.2004.03.017>